

CENG322 - Video Assignment 3

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1. What are the memory waste problems of KV cache management in previous systems? How can you correlate this waste with the concepts covered in our lectures?

In previous systems serving large language models (LLM), memory wastage occurs due to inefficient management of KV memory. The KV cache stores the key-value vectors. As new tokens are generated, the KV cache grows and takes up more space. These systems typically allocate a large chunk of memory for each request. However, most requests do not need this much space. Therefore, a large portion of the allocated memory goes unused. This results in two main types of fragmentation:

- Internal Fragmentation: The allocated memory exceeds the actual space required for requests because most sequences do not reach the maximum length. The unused part of the allocated memory remains wasted.
- External Fragmentation: Since each request may have different maximum lengths, memory is fragmented across the GPU, making it difficult to allocate large continuous memory for new requests.

Relation to Lecture Topics:

- This problem is related to the memory allocation and fragmentation issues we saw in class.
- Virtual memory concepts allow for better utilization of physical memory. Non-contiguous allocation and paging help solve these problems.

2. What does PagedAttention (virtualized KV cache) offer to solve the memory waste problems? How can you correlate this solution with the concepts covered in our lectures?

Instead of allocating a large block of memory, it divides the KV cache into smaller, fixed-size blocks. Each block can store a small number of token states, which helps prevent both internal and external fragmentation.

- Virtual addressing: The system maintains a mapping table that links logical blocks to physical blocks. This allows blocks to be stored in any available space, rather than side by side.

With this method, PagedAttention uses more efficient memory than older systems.

3. After watching the video, do you have the motivation to read the paper with more technical details and results? Why or why not?

Yes, I was motivated to read the paper. The video nicely summarized how PagedAttention works, but when I looked at the paper, I saw that there were more technical details and graphics. For example, the experimental results supporting the structure of the block table and how they set the block size are available in the paper. Although most of the technical parts are difficult to understand, the parts not mentioned in the video are nicely supported by graphics.

4. After watching the video, do you have the motivation to look at the vLLM code? Why or why not?

Yes, I was motivated to explore the vLLM code. I saw that vLLM was open source on GitHub and I also heard that it was built on frameworks I knew, such as PyTorch and HuggingFace. Examining the open source code can help me better understand how this type of memory management is implemented in practice. It also gives me a chance to see how theoretical operating system concepts we have seen in class are implemented in the field of machine learning.